

Engineering Mathematics

Previous Year Questions • 1993–2026

VECTOR CALCULUS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Vector Calculus (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Vector Calculus

Topics

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2. Cylindrical and Spherical Coordinate Systems
3. Gradient, Divergence and Curl
4. Line Integrals and Surface Integrals
5. Green's Theorem
6. Stoke's Theorem and Gauss Divergence Theorem

Note to Reader

Vector Calculus is present in branches with paper codes CE, ME, EE, EC, IN, CH, PI, AE, AG, MT, MN according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 28 **MCQ:** 18 **MSQ:** 2 **NAT:** 8
1M: 12 **2M:** 16 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2023)

EE — Electrical Engineering

Questions: 27 **MCQ:** 18 **MSQ:** 0 **NAT:** 9
1M: 10 **2M:** 17 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–6M(2022)

IN — Instrumentation Engineering

Questions: 11 **MCQ:** 9 **MSQ:** 0 **NAT:** 2
1M: 7 **2M:** 4 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–2M(2025)

ME — Mechanical Engineering

Questions: 44 **MCQ:** 32 **MSQ:** 0 **NAT:** 12
1M: 20 **2M:** 24 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–5M(2025)

CE — Civil Engineering

Questions: 17 **MCQ:** 11 **MSQ:** 2 **NAT:** 4
1M: 6 **2M:** 11 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–2M(2024)

CH — Chemical Engineering

Questions: 6 **MCQ:** 4 **MSQ:** 0 **NAT:** 2
1M: 2 **2M:** 4 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2023)–3M(2026)

PI — Production & Industrial Engineering

Questions: 12 **MCQ:** 9 **MSQ:** 0 **NAT:** 3
1M: 9 **2M:** 3 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–3M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q4.26.1 MCQ 2026 1M

A surface is given by $z^2 = 2x^2 - y^2$ and $\vec{n}$ and $-\vec{n}$ are unit normal vectors to the surface at the point $\vec{P} = \hat{i} + \sqrt{2}\,\hat{k}$. Which of the following vectors can be $\vec{n}$, where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vectors along x, y and z axes, respectively?

A. $\hat{i} - \sqrt{2}\,\hat{k}$

B. $\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k}$

C. $\sqrt{2}\hat{i} - \sqrt{3}\hat{k}$

D. $\frac{\sqrt{2}\hat{i} - \hat{k}}{\sqrt{3}}$

Q4.24.1 MSQ 2024 1M

Let $\rho(x, y, z, t)$ and $u(x, y, z, t)$ represent density and velocity, respectively, at a point (x, y, z) and time t. Assume $\frac{\partial \rho}{\partial t}$ is continuous. Let V be an arbitrary volume in space enclosed by the closed surface S and $\hat{n}$ be the outward unit normal of S.

Which of the following equations is/are equivalent to $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$?

☐ **A.** $\int_V \frac{\partial \rho}{\partial t}\, dv = -\oint_S \rho u \cdot \hat{n}\, ds$

☐ **B.** $\int_V \frac{\partial \rho}{\partial t}\, dv = \oint_S \rho u \cdot \hat{n}\, ds$

☐ **C.** $\int_V \frac{\partial \rho}{\partial t}\, dv = -\int_V \nabla \cdot (\rho u)\, dv$

☐ **D.** $\int_V \frac{\partial \rho}{\partial t}\, dv = \int_V \nabla \cdot (\rho u)\, dv$

Q4.24.2 MSQ 2024 2M

Let F_1, F_2, and F_3 be functions of (x, y, z). Suppose that for every given pair of points A and B in space, the line integral $\int_C (F_1 dx + F_2 dy + F_3 dz)$ evaluates to the same value along any path C that starts at A and ends at B. Then which of the following is/are true?

☐ **A.** For every closed path Γ, we have $\oint_\Gamma (F_1 dx + F_2 dy + F_3 dz) = 0$.

☐ **B.** There exists a differentiable scalar function $f(x, y, z)$ such that

$$F_1 = \frac{\partial f}{\partial x},\ F_2 = \frac{\partial f}{\partial y},\ F_3 = \frac{\partial f}{\partial z}.$$

☐ **C.** $\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0.$

☐ **D.** $\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z},\ \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x},\ \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}.$

Q4.23.1 MCQ 2023 1M

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ be two vectors. The value of the coefficient α in the expression $\mathbf{v}_1 = \alpha \mathbf{v}_2 + \mathbf{e}$, which minimizes the length of the error vector $\mathbf{e}$, is

A. $\frac{7}{2}$ **B.** $-\frac{2}{7}$ **C.** $\frac{2}{7}$ **D.** $-\frac{7}{2}$

Q4.23.2 MCQ 2023 1M

The rate of increase, of a scalar field $f(x, y, z) = xyz$, in the direction $\mathbf{v} = (2, 1, 2)$ at a point $(0, 2, 1)$ is

A. $\frac{2}{3}$ **B.** $\frac{4}{3}$ **C.** 2 **D.** 4

Q4.23.3 MCQ 2023 2M

The value of the line integral

$$\int_P^Q \left(z^2\, dx + 3y^2\, dy + 2xz\, dz\right)$$

along the straight line joining the points $P(1, 1, 2)$ and $Q(2, 3, 1)$ is

A. 20 **B.** 24 **C.** 29 **D.** -5

Q4.22.1 MCQ 2022 1M

Consider the two-dimensional vector field $\vec{F}(x, y) = x\,\vec{i} + y\,\vec{j}$, where $\vec{i}$ and $\vec{j}$ denote the unit vectors along the x-axis and the y-axis, respectively. A contour C in the x-y plane, as shown in the figure, is composed of two horizontal lines connected at the two ends by two semicircular arcs of unit radius. The contour is traversed in the counter-clockwise sense. The value of the closed path integral

$$\oint_C \vec{F}(x, y) \cdot (dx\,\vec{i} + dy\,\vec{j})$$

is _____.

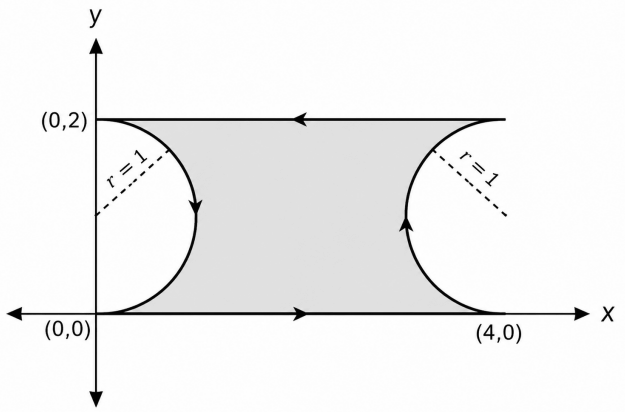


Fig. Q4.22.1

A. 0 **B.** 1 **C.** $8 + 2\pi$ **D.** -1

Q4.21.1 MCQ 2021 1M

The vector function $\mathbf{F}(\mathbf{r}) = -x\,\hat{i} + y\,\hat{j}$ is defined over a circular arc C shown in the figure.

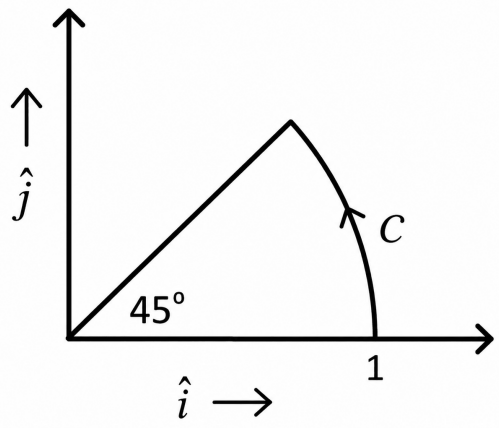


Fig. Q4.21.1

The line integral of $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$ is

A. $\frac{1}{2}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{6}$ **D.** $\frac{1}{3}$

Q4.20.1 MCQ 2020 1M

For a vector field $\vec{A}$, which one of the following is FALSE?

A. $\vec{A}$ is solenoidal if $\nabla \cdot \vec{A} = 0$.

B. $\nabla \times \vec{A}$ is another vector field.

C. $\vec{A}$ is irrotational if $\nabla^2 \vec{A} = 0$.

D. $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$.

Q4.19.1 MCQ 2019 2M

Consider the line integral

$$\int_C (x\,dy - y\,dx)$$

the integral being taken in a counterclockwise direction over the closed curve C that forms the boundary of the region R shown in the figure below. The region R is the area enclosed by the union of a 2×3 rectangle and a semi-circle of radius 1. The line integral evaluates to

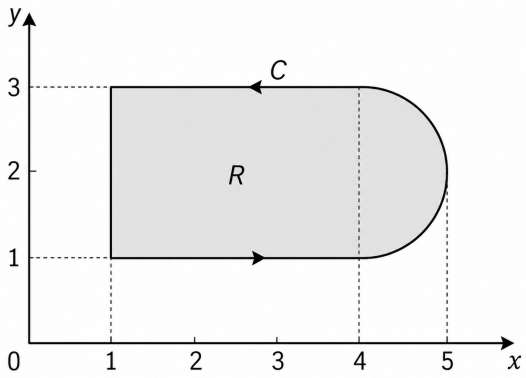


Fig. Q4.19.1

A. $6 + \pi/2$ **B.** $8 + \pi$ **C.** $12 + \pi$ **D.** $16 + 2\pi$

Q4.17.1 NAT 2017 2M

Let $I = \int_C (2z\,dx + 2y\,dy + 2x\,dz)$ where x, y, z are real, and let C be the straight line segment from point $A : (0, 2, 1)$ to point $B : (4, 1, -1)$. The value of I is _____.

Q4.17.2 MCQ 2017 2M

If the vector function

$$\vec{F} = \hat{a}_x(3y - k_1 z) + \hat{a}_y(k_2 x - 2z) - \hat{a}_z(k_3 y + z)$$

is irrotational, then the values of the constants k_1, k_2 and k_3, respectively, are

A. 0.3, -2.5, 0.5 **B.** 0.0, 3.0, 2.0

C. 0.3, 0.33, 0.5 **D.** 4.0, 3.0, 2.0

Q4.17.3 NAT 2017 1M

The smaller angle (in degrees) between the planes $x+y+z = 1$ and $2x - y + 2z = 0$ is _____.

Q4.16.1 NAT 2016 2M

The integral $\frac{1}{2\pi}\iint_D (x + y + 10)\, dx\, dy$, where D denotes the disc: $x^2 + y^2 \le 4$ evaluates to _____.

Q4.16.2 NAT 2016 2M

Suppose C is the closed curve defined as the circle $x^2 + y^2 = 1$ with C oriented anti-clockwise. The value of $\oint (xy^2\, dx + x^2 y\, dy)$ over the curve C equals _____.

Q4.15.1 MCQ 2015 2M

A vector $\vec{P}$ is given by

$$\vec{P} = x^3 y\, \hat{a}_x - x^2 y^2\, \hat{a}_y - x^2 yz\, \hat{a}_z$$

Which of the following statements is TRUE?

A. $\vec{P}$ is solenoidal but not irrotational.

B. $\vec{P}$ is irrotational but not solenoidal.

C. $\vec{P}$ is neither solenoidal nor irrotational.

D. $\vec{P}$ is both solenoidal and irrotational.

Q4.14.1 NAT 2014 2M

If $\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$, and $|\vec{r}| = r$, then

$$\text{div}\left(r^2 \nabla(\ln r)\right) = _____.$$

Q4.14.2 NAT 2014 1M

The directional derivative of

$$f(x, y) = \frac{xy}{\sqrt{2}}(x + y)$$

at $(1, 1)$ in the direction of the unit vector at an angle of $\frac{\pi}{4}$ with y-axis, is given by _____.

Q4.14.3 NAT 2014 2M

Given $\vec{F} = z\hat{a}_x + x\hat{a}_y + y\hat{a}_z$. If S represents the portion of the sphere

$$x^2 + y^2 + z^2 = 1$$

for $z \ge 0$, then

$$\int_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

is _____.

Q4.13.1 MCQ 2013 1M

Consider a vector field $\vec{A}(\vec{r})$. The closed loop line integral $\oint \vec{A} \bullet d\vec{l}$ can be expressed as

A. $\oiint (\nabla \times \vec{A}) \bullet d\vec{s}$ over the closed surface bounded by the loop

B. $\iiint (\nabla \bullet \vec{A})\, dv$ over the closed volume bounded by the loop

C. $\iiint (\nabla \bullet \vec{A})\, dv$ over the open volume bounded by the loop

D. $\iint (\nabla \times \vec{A}) \bullet d\vec{s}$ over the open surface bounded by the loop

Q4.13.2 MCQ 2013 1M

The divergence of the vector field $\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ is

A. 0 **B.** 1/3 **C.** 1 **D.** 3

Q4.12.1 MCQ 2012 2M

The direction of vector $\vec{A}$ is radially outward from the origin, with $|\vec{A}| = Kr^n$, where $r^2 = x^2 + y^2 + z^2$ and K is constant. The value of n for which $\nabla \cdot \vec{A} = 0$ is

A. -2 **B.** 2

C. 1 **D.** 0

Q4.11.1 MCQ 2011 2M

Consider a closed surface S surrounding a volume V. If $\vec{r}$ is the position vector of a point inside S, with $\hat{n}$ the unit normal on S, the value of the integral

$$\oiint_S 5\vec{r} \cdot \hat{n}\, ds$$

is

A. 3V **B.** 5V

C. 10V **D.** 15V

Q4.10.1 MCQ 2010 2M

If $\vec{A} = xy\,\hat{a}_x + x^2\,\hat{a}_y$, then $\oint_C \vec{A} \bullet d\vec{l}$ over the path shown in the figure is

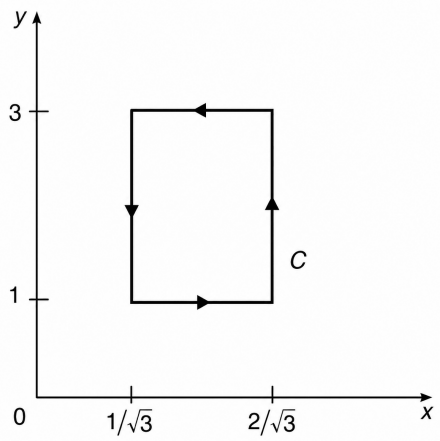


Fig. Q4.10.1

A. 0 **B.** $\frac{2}{\sqrt{3}}$ **C.** 1 **D.** $2\sqrt{3}$

Q4.08.1 MCQ 2008 2M

Consider points P and Q in the x-y plane, with $P = (1, 0)$ and $Q = (0, 1)$. The line integral

$$2\int_P^Q (x\,dx + y\,dy)$$

along the semicircle with the line segment PQ as its diameter

A. is -1

B. is 0

C. is 1

D. depends on the direction (clockwise or anti-clockwise) of the semicircle

Q4.08.2 MCQ 2008 2M

The value of the integral of the function $g(x, y) = 4x^3 + 10y^4$ along the straight line segment from the point $(0, 0)$ to the point $(1, 2)$ in the x-y plane is

A. 33 **B.** 35 **C.** 40 **D.** 56

Q4.06.1 MCQ 2006 1M

$\iint (\nabla \times P) \bullet d\vec{s}$, where P is a vector, is equal to

A. $\oint P \bullet d\vec{l}$ **B.** $\oint \nabla \times \nabla \times P \bullet d\vec{l}$

C. $\oint \nabla \times P \bullet d\vec{l}$ **D.** $\iiint \nabla \bullet P\, dv$

Q4.93.1 NAT 1993 2M

Given, $\vec{V} = x\cos^2 y\,\hat{\imath} + x^2 e^z\,\hat{\jmath} + z\sin^2 y\,\hat{k}$ and S the surface of a unit cube with one corner at the origin and edges parallel to the coordinate axes, the value of the integral $\iint_S \vec{V} \cdot \hat{n}\, dS$ is _____

Electrical Engineering

Q4.26.1 NAT 2026 1M

Given that $\vec{F}(x, y, z) = \sin(y)\,\hat{x} + \cos(x)\,\hat{y} + 5\,\hat{z}$, the integral

$$\oiint_S \vec{F}(x, y, z) \cdot d\vec{s}$$

over the unit sphere S centered at the origin evaluates to _____ (Round off to one decimal place)

Q4.25.1 NAT 2025 2M

Let $(x, y) \in \mathbb{R}^2$. The rate of change of the real valued function,

$$V(x, y) = x^2 + x + y^2 + 1$$

at the origin in the direction of the point $(1, 2)$ is _____ (round off to the nearest integer).

Q4.24.1 MCQ 2024 2M

Consider a vector $\vec{u} = 2\hat{x} + \hat{y} + 2\hat{z}$, where $\hat{x}, \hat{y}, \hat{z}$ represent unit vector along the coordinate axes x, y, z respectively. The directional derivative of the function

$$f(x, y, z) = 2\ln(xy) + \ln(yz) + 3\ln(xz)$$

at the point $(x, y, z) = (1, 1, 1)$ in the direction of $\vec{u}$ is

A. 0 **B.** $\frac{7}{5\sqrt{2}}$

C. 7 **D.** 21

Q4.23.1 NAT 2023 2M ○○○

A quadratic function of two variables is given as

$$f(x_1, x_2) = x_1^2 + 2x_2^2 + 3x_1 + 3x_2 + x_1x_2 + 1$$

The magnitude of the maximum rate of change of the function at the point $(1, 1)$ is _____(Round off to the nearest integer).

Q4.23.2 NAT 2023 2M ○○○

The closed curve shown in the figure is described by $r = 1+\cos\theta$, where $r = \sqrt{x^2 + y^2}$; $x = r\cos\theta$, $y = r\sin\theta$. The magnitude of the line integral of the vector field $\mathbf{F} = -y\hat{i} + x\hat{j}$ around the closed curve is _____ (Round off to 2 decimal places).

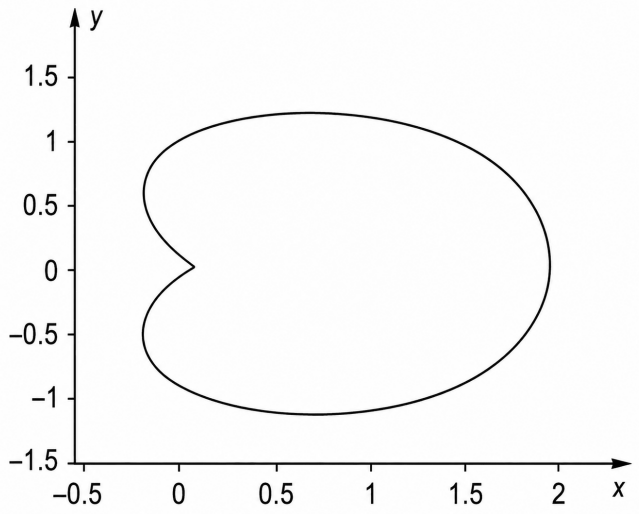


Fig. Q4.23.2

Q4.23.3 MCQ 2023 1M ○○○

For a given vector $W = [1, 2, 3]^T$, the vector normal to the plane defined by $W^T X = 1$ is

A. $[-2, -2, 2]^T$ **B.** $[3, 0, -1]^T$

C. $[3, 2, 1]^T$ **D.** $[1, 2, 3]^T$

Q4.22.1 MCQ 2022 2M ○○○

Let R be a region in the first quadrant of the xy plane enclosed by a closed curve C considered in counter-clockwise direction. Which of the following expressions does not represent the area of the region R?

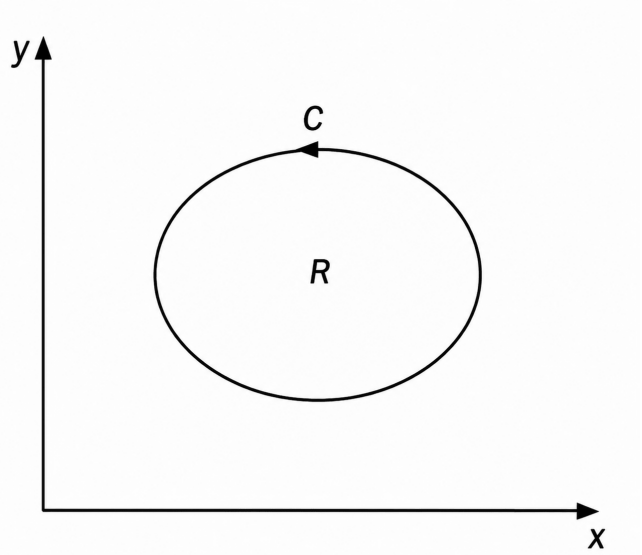


Fig. Q4.22.1

A. $\iint_R dx\,dy$ **B.** $\oint_C x\,dy$

C. $\oint_C y\,dx$ **D.** $\frac{1}{2}\oint_C (x\,dy - y\,dx)$

Q4.22.2 MCQ 2022 2M ○○○

Let $\vec{E}(x, y, z) = 2x^2\hat{i} + 5y\hat{j} + 3z\hat{k}$. The value of

$$\iiint_V (\vec{\nabla} \cdot \vec{E})\, dV,$$

where V is the volume enclosed by the unit cube defined by $0 \le x \le 1$, $0 \le y \le 1$, and $0 \le z \le 1$, is

A. 3 **B.** 8

C. 10 **D.** 5

Q4.22.3 MCQ 2022 2M ○○○

Let $f(x, y, z) = 4x^2 + 7xy + 3xz^2$. The direction in which the function $f(x, y, z)$ increases most rapidly at point $P = (1, 0, 2)$ is

A. $20\hat{i} + 7\hat{j}$ **B.** $20\hat{i} + 7\hat{j} + 12\hat{k}$

C. $20\hat{i} + 12\hat{k}$ **D.** $20\hat{i}$

Q4.20.1 NAT 2020 2M ○○○

Let a_x and a_y be unit vectors along x and y directions, respectively. A vector function is given by $F = a_x y - a_y x$. The line integral of the above function $\int_C F \cdot dl$ along the curve C, which follows the parabola $y = x^2$, as shown below is _____ (rounded off to 2 decimal places).

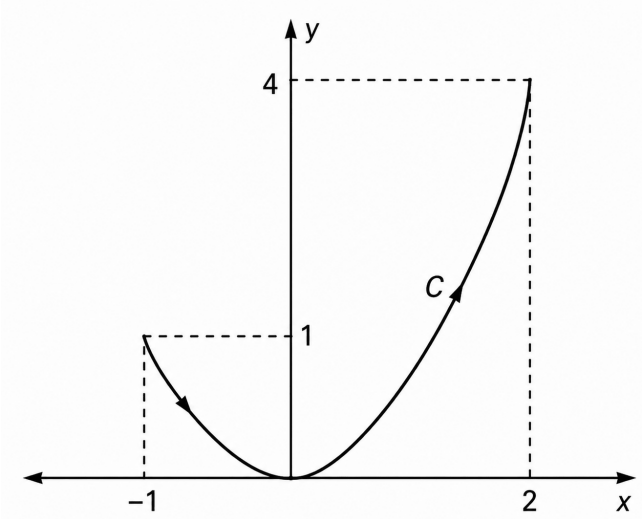


Fig. Q4.20.1

Q4.20.2 MCQ 2020 2M

The vector function expressed by

$$F = a_x(5y - k_1 z) + a_y(3z + k_2 x) + a_z(k_3 y - 4x)$$

represents a conservative field, where a_x, a_y, a_z are unit vectors along x, y and z directions, respectively. The values of constants k_1, k_2, k_3 are given by:

A. $k_1 = 3,\ k_2 = 3,\ k_3 = 7$ **B.** $k_1 = 4,\ k_2 = 5,\ k_3 = 3$

C. $k_1 = 3,\ k_2 = 8,\ k_3 = 5$ **D.** $k_1 = 0,\ k_2 = 0,\ k_3 = 0$

Q4.19.1 NAT 2019 1M

If $f = 2x^3 + 3y^2 + 4z$, the value of line integral $\int_C \nabla f \cdot d\mathbf{r}$ evaluated over contour C formed by the segments $(-3, -3, 2) \rightarrow (2, -3, 2) \rightarrow (2, 6, 2) \rightarrow (2, 6, -1)$ is _____.

Q4.18.1 NAT 2018 2M

As shown in the figure, C is the arc from the point $(3, 0)$ to the point $(0, 3)$ on the circle $x^2 + y^2 = 9$. The value of the integral

$$\int_C \left[(y^2 + 2yx)\,dx + (2xy + x^2)\,dy\right]$$

is _____ (upto 2 decimal places).

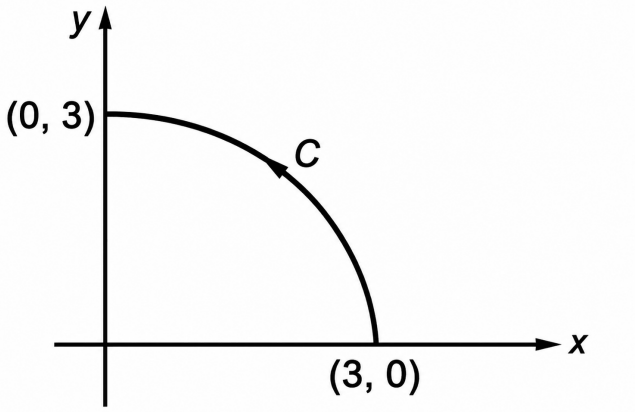


Fig. Q4.18.1

Q4.18.2 MCQ 2018 1M

The value of the directional derivative of the function $\phi(x, y, z) = xy^2 + yz^2 + zx^2$ at the point $(2, -1, 1)$ in the direction of the vector $\mathbf{p} = \hat{i} + 2\hat{j} + 2\hat{k}$ is

A. 1 **B.** 0.95

C. 0.93 **D.** 0.9

Q4.16.1 NAT 2016 2M

The line integral of the vector field $F = 5xz\hat{i} + (3x^2 + 2y)\hat{j} + x^2 z\hat{k}$ along a path from $(0, 0, 0)$ to $(1, 1, 1)$ parameterized by (t, t^2, t) is _____.

Q4.16.2 MCQ 2016 1M

The value of the line integral

$$\int_C (2xy^2\,dx + 2x^2 y\,dy + dz)$$

along a path joining the origin $(0, 0, 0)$ and the point $(1, 1, 1)$ is

A. 0 **B.** 2 **C.** 4 **D.** 6

Q4.15.1 NAT 2015 2M

The volume enclosed by the surface $f(x, y) = e^x$ over the triangle bounded by the lines $x = y$, $x = 0$ and $y = 1$ in the xy plane is _____.

Q4.14.1 MCQ 2014 1M

Let $\nabla \cdot (f\vec{v}) = x^2 y + y^2 z + z^2 x$, where f and $\vec{v}$ are scalar and vector fields respectively. If $\vec{v} = y\hat{i} + z\hat{j} + x\hat{k}$, then $\vec{v} \cdot \nabla f$ is

A. $x^2 y + y^2 z + z^2 x$ **B.** $2xy + 2yz + 2zx$

C. $x + y + z$ **D.** 0

Q4.14.2 MCQ 2014 2M

The line integral of function $F = yz\hat{i}$, in the counter clockwise direction, along the circle $x^2 + y^2 = 1$ at $z = 1$ is

A. -2π **B.** $-\pi$

C. π **D.** 2π

Q4.13.1 MCQ 2013 1M

Given a vector field $\vec{F} = y^2x\hat{a}_x - yz\hat{a}_y - x^2\hat{a}_z$, the line integral $\int \vec{F} \cdot d\vec{l}$ evaluated along a segment on the x-axis from $x = 1$ to $x = 2$ is

A. -2.33 **B.** 0

C. 2.33 **D.** 7

Q4.13.2 MCQ 2013 1M

The curl of the gradient of the scalar field defined by $V = 2x^2y + 3y^2z + 4z^2x$ is

A. $4xy\,a_x + 6yz\,a_y + 8zx\,a_z$

B. $4a_x + 6a_y + 8a_z$

C. $(4xy + 4z^2)\,a_x + (2x^2 + 6yz)\,a_y + (3y^2 + 8zx)\,a_z$

D. 0

Q4.12.1 MCQ 2012 2M

The direction of vector $\vec{A}$ is radially outward from the origin, with $\left|\vec{A}\right| = kr^n$, where $r^2 = x^2 + y^2 + z^2$ and k is constant. The value of n for which $\nabla \cdot \vec{A} = 0$ is

A. -2 **B.** 2

C. 1 **D.** 0

Q4.11.1 MCQ 2011 1M

The two vectors $[1, 1, 1]$ and $[1, a, a^2]$ where $a = \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$, are

A. orthonormal. **B.** orthogonal.

C. parallel. **D.** collinear.

Q4.10.1 MCQ 2010 1M

Divergence of the three-dimensional radial vector field $\vec{r}$ is

A. 3 **B.** $1/r$

C. $\hat{i} + \hat{j} + \hat{k}$ **D.** $3(\hat{i} + \hat{j} + \hat{k})$

Q4.09.1 MCQ 2009 2M

$\vec{F}(x, y) = (x^2 + xy)\hat{a}_x + (y^2 + xy)\hat{a}_y$. Its line integral over the straight line from $(x, y) = (0, 2)$ to $(x, y) = (2, 0)$ evaluates to

A. -8 **B.** 4

C. 8 **D.** 0

Q4.06.1 MCQ 2006 2M

A surface $S(x, y) = 2x+5y-3$ is integrated once over a path consisting of the points that satisfy $(x+1)^2+(y-1)^2 = \sqrt{2}$. The integral evaluates to

A. $17\sqrt{2}$ **B.** $\frac{17}{\sqrt{2}}$

C. $\frac{\sqrt{2}}{17}$ **D.** 0

Q4.05.1 MCQ 2005 2M

For the scalar field $u = \frac{x^2}{2} + \frac{y^2}{3}$, magnitude of the gradient at the point $(1, 3)$ is

A. $\sqrt{\frac{13}{9}}$ **B.** $\sqrt{\frac{9}{2}}$

C. $\sqrt{5}$ **D.** $\frac{9}{2}$

Instrumentation Engineering

Q4.25.1 MCQ 2025 2M

The value of the surface integral

$$\iint_S [(2x + z)\,dy\,dz + (2x + z)\,dx\,dz + (2z + y)\,dx\,dy]$$

over the sphere $S : x^2 + y^2 + z^2 = 9$ is

A. 72π **B.** 144π

C. 36π **D.** 432π

Q4.20.1 MCQ 2020 1M

The unit vectors along the mutually perpendicular x, y and z axes are $\hat{i}$, $\hat{j}$ and $\hat{k}$ respectively.Consider the plane $z = 0$ and two vector $\vec{a}$ and $\vec{b}$ on that plane such that $\vec{a} \neq \alpha\vec{b}$ for any scalar α. A vector perpendicular to both $\vec{a}$ and $\vec{b}$ is

A. $\hat{i} - \hat{j}$ **B.** $-\hat{j}$

C. $\hat{k}$ **D.** $\hat{i}$

Q4.19.1 MCQ 2019 1M

$\vec{a}, \vec{b}, \vec{c}$ are three orthogonal vectors, Given that $\vec{a} = \hat{i} + 2\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, the vector $\vec{c}$ is parallel to

A. $\hat{i} + 2\hat{j} + 3\hat{k}$ **B.** $2\hat{i} + \hat{j}$

C. $2\hat{i} - \hat{j}$ **D.** $4\hat{k}$

Q4.18.1 MCQ 2018 2M

Given $\vec{F} = (x^2 - 2y)\,\hat{i} - 4yz\,\hat{j} + 4xz^2\hat{k}$, the value of the line integral

$$\int_C \vec{F} \cdot d\vec{l}$$

along the straight line C from $(0, 0, 0)$ to $(1, 1, 1)$ is

A. $\frac{3}{16}$ **B.** 0

C. $-\frac{5}{12}$ **D.** -1

Q4.17.1 NAT 2017 2M

The angle between two vectors $X_1 = [2\ 6\ 14]^T$ and $X_2 = [-12\ 8\ 16]^T$ in radian is _____.

Q4.16.1 MCQ 2016 1M

The vector that is NOT perpendicular to the vectors $(i + j + k)$ and $(i + 2j + 3k)$ is _____.

A. $(i - 2j + k)$ **B.** $(-i + 2j - k)$

C. $(0i + 0j + 0k)$ **D.** $(4i + 3j + 5k)$

Q4.15.1 MCQ 2015 1M

The magnitude of the directional derivative of the function $f(x, y) = x^2 + 3y^2$ in a direction normal to the circle $x^2 + y^2 = 2$, at the point $(1, 1)$, is

A. $4\sqrt{2}$ **B.** $5\sqrt{2}$

C. $7\sqrt{2}$ **D.** $9\sqrt{2}$

Q4.14.1 NAT 2014 1M

A vector is defined as

$$f = y\hat{i} + x\hat{j} + z\hat{k}$$

where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors in Cartesian (x, y, z) coordinate system.The surface integral $\oiint f \cdot ds$ over the closed surface S of a cube with vertices having the following coordinates: $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(1, 0, 1)$, $(1, 1, 1)$, $(0, 1, 1)$ is _____

Q4.13.1 MCQ 2013 1M

For a vector E, which one of the following statements is NOT TRUE?

A. If $\nabla \cdot E = 0$, E is called solenoidal.

B. If $\nabla \times E = 0$, E is called conservative.

C. If $\nabla \times E = 0$, E is called irrotational.

D. If $\nabla \cdot E = 0$, E is called irrotational.

Q4.12.1 MCQ 2012 2M

The direction of vector A is radially outward from origin, with $|A| = kr^n$ where $r^2 = x^2 + y^2 + z^2$ and k is a constant. The value of n for which $\nabla \cdot A = 0$ is

A. -2 **B.** 1

C. 2 **D.** 0

Q4.09.1 MCQ 2009 1M

A sphere of unit radius is centered at the origin. The unit normal at a point (x, y, z) on the surface of the sphere is the vector

A. (x, y, z) **B.** $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

C. $\left(\frac{x}{\sqrt{3}}, \frac{y}{\sqrt{3}}, \frac{z}{\sqrt{3}}\right)$ **D.** $\left(\frac{x}{\sqrt{2}}, \frac{y}{\sqrt{2}}, \frac{z}{\sqrt{2}}\right)$

Mechanical Engineering

Q4.26.1 MCQ 2026 1M

Domain A is bounded by curve $x^2 = 4y$, ordinate $x = 2$, and x axis. The value of $\iint_A y\,dxdy$ is

A. $\frac{1}{5}$ **B.** $\frac{1}{3}$

C. $\frac{5}{12}$ **D.** $\frac{1}{2}$

Q4.25.1 NAT 2025 2M

Consider a velocity field $\vec{V} = 3z\hat{i} + 0\hat{j} + Cx\hat{k}$, where C is a constant. if the flow is irrotational, the value of C is _____ (rounded off to 1 decimal place).

Q4.25.2 NAT 2025 2M

The directional derivative of the function f given below at the point (1, 0) in the direction of $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ is _____ (Rounded off to 1 decimal place).

$$f(x, y) = x^2 + xy^2$$

Q4.25.3 MCQ 2025 1M

The divergence of the curl of a vector field is

A. the magnitude of this vector field

B. the argument of this vector field

C. the magnitude of the curl of this vector field

D. Zero

Q4.24.1 MCQ 2024 1M

The value of the surface integral

$$\oiint_S z\,dxdy$$

where S is the external surface of the sphere $x^2 + y^2 + z^2 = R^2$ is

A. $4\pi R^3$ **B.** $\frac{4\pi R^3}{3}$

C. 0 **D.** πR^3

Q4.23.1 NAT 2023 1M

A vector field

$$B(x, y, z) = x\hat{i} + y\hat{j} - 2z\hat{k}$$

is defined over a conical region having height $h = 2$, base radius $r = 3$ and axis along z, as shown in the figure. The base of the cone lies in the x-y plane and is centered at the origin. If n denotes the unit outward normal to the curved surface S of the cone, the value of the integral,

$$\int_S B \cdot n\,dS$$

equals _____. [Answer in integer]

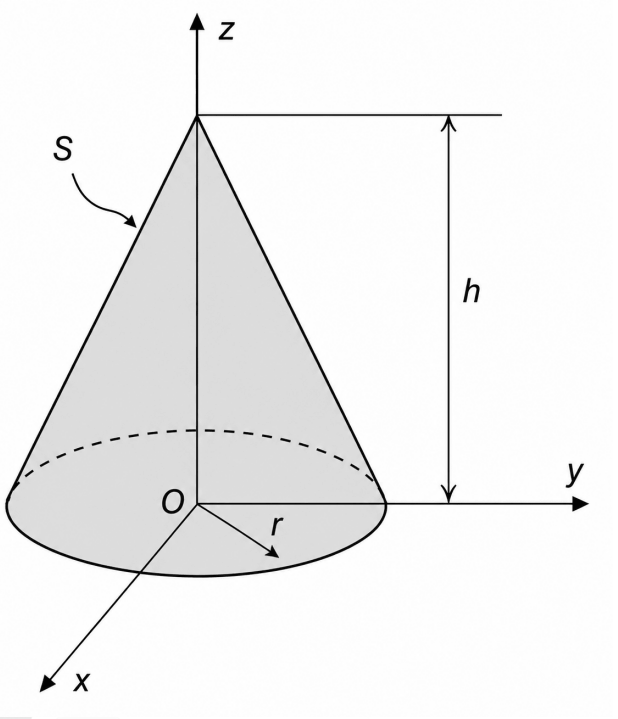


Fig. Q4.23.1

Q4.22.1 MCQ 2022 1M

Consider a cube of unit edge length and sides parallel to coordinate axes with its centroid at the point (1, 2, 3). The surface integral $\int_A \vec{F} \cdot d\vec{A}$ of a vector field, $\vec{F} = 3x\hat{i} + 5y\hat{j} + 6z\hat{k}$ over the entire surface A area of the cube is _____.

A. 14 **B.** 27

C. 28 **D.** 31

Q4.22.2 NAT 2022 2M

Consider two vectors:

$$\vec{a} = 5i + 7j + 2k$$

$$\vec{b} = 3i - j + 6k$$

Magnitude of the component of $\vec{a}$ orthogonal to $\vec{b}$ in the plane containing the vectors $\vec{a}$ and $\vec{b}$ is _____ (round off to 2 decimal places).

Q4.22.3 MCQ 2022 1M

Given a function $\phi = \frac{1}{2}(x^2 + y^2 + z^2)$ in three-dimensional Cartesian space, the value of the surface integral

$$\oiint_S \hat{n} \cdot \nabla\phi\,dS,$$

where S is the surface of a sphere of unit radius and $\hat{n}$ is the outward unit normal vector on S, is

A. 4π **B.** 3π

C. $\frac{4\pi}{3}$ **D.** 0

Q4.20.1 MCQ 2020 2M

The directional derivative $f(x,\ y,\ z) = xyz$ at point $(-1,\ 1,\ 3)$ in the direction of vector $\hat{i} - 2\hat{j} + 2\hat{k}$ is

A. $\frac{7}{3}$ **B.** 7

C. $-\frac{7}{3}$ **D.** $3\hat{i} - 3\hat{j} - \hat{k}$

Q4.20.2 MCQ 2020 2M

A vector field is defined as

$$\vec{F}(x,y,z) = \frac{x}{[x^2+y^2+z^2]^{3/2}}\hat{i} + \frac{y}{[x^2+y^2+z^2]^{3/2}}\hat{j} + \frac{z}{[x^2+y^2+z^2]^{3/2}}\hat{k}$$

where, $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the axes of a right-handed rectangular/Cartesian coordinate system. The surface integral $\iint \vec{F} d\vec{S}$ (where $d\vec{S}$ is an elemental surface area vector) evaluated over the inner and outer surfaces of a spherical shell formed by two concentric spheres with origin as the center, and internal and external radii of 1 and 2, respectively, is

A. 0 **B.** 2π

C. 8π **D.** 4π

Q4.20.3 NAT 2020 1M

For three vectors $\vec{A} = 2\hat{j} - 3\hat{k}$, $\vec{B} = -2\hat{i} + \hat{k}$ and $\vec{C} = 3\hat{i} - \hat{j}$, where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along the axes of a right-handed rectangular/Cartesian coordinate system, the value of

$$\left(\vec{A} \cdot (\vec{B} \times \vec{C}) + 6\right)$$

is _____.

Q4.19.1 MCQ 2019 2M

Given a vector $\vec{U} = \frac{1}{3}\left(-y^3\hat{i} + x^3\hat{j} + z^3\hat{k}\right)$ and $\hat{n}$ as the unit normal vector to the surface of the hemisphere $(x^2 + y^2 + z^2 = 1;\ z \geq 0)$, the value of integral $\int (\nabla \times \vec{U}) \cdot \hat{n}\, dS$ evaluated on the curved surface of the hemisphere S is

A. π **B.** $\frac{\pi}{2}$

C. $-\frac{\pi}{2}$ **D.** $\frac{\pi}{3}$

Q4.19.2 MCQ 2019 1M

The directional derivative of the function $f(x, y) = x^2 + y^2$ along a line directed from $(0, 0)$ to $(1, 1)$, evaluated at point $x = 1,\ y = 1$ is

A. $4\sqrt{2}$ **B.** 2

C. $2\sqrt{2}$ **D.** $\sqrt{2}$

Q4.19.3 MCQ 2019 1M

The position vector OP of point $P(20, 10)$ is rotated anti-clockwise in $X - Y$ plane by an angle of $30°$ such that the point P occupies position Q as shown in the figure. The coordinates (x, y) of Q are

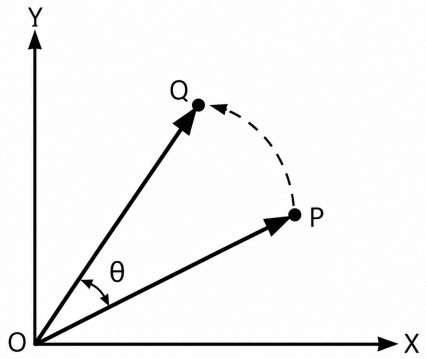


Fig. Q4.19.3

A. $(13.40, 22.32)$ **B.** $(22.32, 8.26)$

C. $(12.32, 18.66)$ **D.** $(18.66, 12.32)$

Q4.18.1 MCQ 2018 2M

For a position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ the norm of the vector can be defined as

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}.$$

Given a function $\phi = \ln|\vec{r}|$, its gradient $\nabla\phi$ is

A. $\vec{r}$ **B.** $\frac{\vec{r}}{|\vec{r}|}$

C. $\frac{\vec{r}}{\vec{r} \cdot \vec{r}}$ **D.** $\frac{\vec{r}}{|\vec{r}|^3}$

Q4.18.2 MCQ 2018 1M

The divergence of the vector field $\vec{u} = e^x(\cos y\, \hat{i} + \sin y\, \hat{j})$ is

A. 0 **B.** $e^x \cos y + e^x \sin y$

C. $2e^x \cos y$ **D.** $2e^x \sin y$

Q4.18.3 MCQ 2018 2M

The value of integral $\oiint_S \vec{r} \cdot \vec{n}\, ds$ over the closed surface S bounding a volume V, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector and $\vec{n}$ is the normal to the surface S, is

A. V **B.** $2V$

C. $3V$ **D.** $4V$

Q4.17.1 NAT 2017 2M

The surface integral $\iint_S \vec{F} \cdot \hat{n}\, dS$ over the surface S of the sphere $x^2 + y^2 + z^2 = 9$, where $\vec{F} = (x + y)\hat{i} + (x + z)\hat{j} + (y + z)\hat{k}$ and $\hat{n}$ is the unit outward surface normal, yields (rounded off to the nearest integer) _____.

Q4.17.2 NAT 2017 2M

For the vector $\vec{V} = 2yz\hat{i} + 3xz\hat{j} + 4xy\hat{k}$, the value of $\nabla \cdot (\nabla \times \vec{V})$ is _____.

Q4.17.3 MCQ 2017 2M

A parametric curve defined by

$$x = \cos\left(\frac{\pi u}{2}\right), \qquad y = \sin\left(\frac{\pi u}{2}\right)$$

in the range $0 \le u \le 1$ is rotated about the x-axis by 360 degrees. Area of the surface generated is

A. $\frac{\pi}{2}$ **B.** π

C. 2π **D.** 4π

Q4.17.4 MCQ 2017 1M

Consider the two-dimensional velocity field given by

$$\vec{V} = (5 + a_1 x + b_1 y)\hat{i} + (4 + a_2 x + b_2 y)\hat{j},$$

where a_1, b_1, a_2 and b_2 are constants. Which one of the following conditions needs to be satisfied for the flow to be incompressible?

A. $a_1 + b_1 = 0$ **B.** $a_1 + b_2 = 0$

C. $a_2 + b_2 = 0$ **D.** $a_2 + b_1 = 0$

Q4.16.1 NAT 2016 2M

The value of the line integral $\oint_C \vec{F} \cdot \vec{r}\, ds$, where C is a circle of radius $\frac{4}{\sqrt{\pi}}$ units is _____.

Here, $\vec{F}(x, y) = y\hat{i} + 2x\hat{j}$ and $\vec{r}$ is the UNIT tangent vector on the curve C at an arc length s from a reference point on the curve. $\hat{i}$ and $\hat{j}$ are the basis vectors in the x-y Cartesian reference. In evaluating the line integral, the curve has to be traversed in the counter-clockwise direction.

Q4.16.2 NAT 2016 2M

A scalar potential ϕ has the following gradient : $\nabla\phi = yz\hat{i} + xz\hat{j} + xy\hat{k}$. Consider the integral $\int_C \nabla\phi \cdot d\vec{r}$ on the curve $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. The curve C is parameterized as follows :

$$\begin{cases} x = t \\ y = t^2 \\ z = 3t^2 \end{cases} \qquad \text{and } 1 \le t \le 3$$

The value of the integral is _____

Q4.15.1 NAT 2015 2M

The surface integral $\iint_S \frac{1}{\pi}(9xi - 3yj) \cdot n\, dS$ over the sphere given by $x^2 + y^2 + z^2 = 9$ is _____.

Q4.15.2 NAT 2015 2M

The value of $\int_C \left[(3x - 8y^2)\, dx + (4y - 6xy)\, dy\right]$, (where C is the boundary of the region boundary by $x = 0$, $y = 0$ and $x + y = 1$) is _____.

Q4.15.3 MCQ 2015 2M

Curl of vector $V(x, y, z) = 2x^2\hat{i} + 3z^2\hat{j} + y^3\hat{k}$ at $x = y = z = 1$ is

A. $-3\hat{i}$ **B.** $3\hat{i}$

C. $3\hat{i} - 4\hat{j}$ **D.** $3\hat{i} - 6\hat{k}$

Q4.15.4 NAT 2015 2M

Consider a spatial curve in three-dimensional space given in parametric form by

$$x(t) = \cos t, \qquad y(t) = \sin t, \qquad z(t) = \frac{2}{\pi}t, 0 \le t \le \frac{\pi}{2}.$$

The length of the curve is _____.

Q4.14.1 MCQ 2014 1M

Divergence of the vector field $x^2z\hat{i} + xy\hat{j} - yz^2\hat{k}$ at $(1, -1, 1)$ is

A. 0 **B.** 3

C. 5 **D.** 6

Q4.14.2 MCQ 2014 1M

Curl of vector $\vec{F} = x^2z^2\hat{i} - 2xy^2z\hat{j} + 2y^2z^3\hat{k}$ is

A. $(4yz^3 + 2xy^2)\hat{i} + 2x^2z\hat{j} - 2y^2z\hat{k}$

B. $(4yz^3 + 2xy^2)\hat{i} - 2x^2z\hat{j} - 2y^2z\hat{k}$

C. $2xz^2\hat{i} - 4xyz\hat{j} + 6y^2z^2\hat{k}$

D. $2xz^2\hat{i} + 4xyz\hat{j} + 6y^2z^2\hat{k}$

Q4.14.3 MCQ 2014 1M

Which one of the following describes the relationship among the three vectors, $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 3\hat{j} + \hat{k}$ and $5\hat{i} + 6\hat{j} + 4\hat{k}$?

A. The vectors are mutually perpendicular

B. The vectors are linearly dependent

C. The vectors are linearly independent

D. The vectors are unit vectors

Q4.14.4 MCQ 2014 2M

The integral

$$\oint_C (y\,dx - x\,dy)$$

is evaluated along the circle $x^2+y^2 = \frac{1}{4}$ traversed in counter clockwise direction. The integral is equal to

A. 0 **B.** $-\frac{\pi}{4}$

C. $-\frac{\pi}{2}$ **D.** $\frac{\pi}{4}$

Q4.13.1 MCQ 2013 2M

The following surface integral is to be evaluated over a sphere for the given steady velocity vector field $F = x\hat{i} + y\hat{j} + z\hat{k}$ defined with respect to a Cartesian coordinate system having $\hat{i}$, $\hat{j}$ and $\hat{k}$ as unit base vectors.

$$\iint_S \frac{1}{4}(F \cdot n)\,dA$$

where S is the sphere, $x^2 + y^2 + z^2 = 1$ and n is the outward unit normal vector to the sphere. The value of the surface integral is

A. π **B.** 2π

C. $3\pi/4$ **D.** 4π

Q4.12.1 MCQ 2012 1M

For the spherical surface $x^2 + y^2 + z^2 = 1$, the unit outward normal vector at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ is given by

A. $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ **B.** $\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$

C. $\hat{k}$ **D.** $\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}$

Q4.10.1 MCQ 2010 2M

Velocity vector of a flow field is given as $\vec{V} = 2xy\hat{i} - x^2z\hat{j}$. The vorticity vector at $(1, 1, 1)$ is

A. $4\hat{i} - \hat{j}$ **B.** $4\hat{i} - \hat{k}$

C. $\hat{i} - 4\hat{j}$ **D.** $\hat{i} - 4\hat{k}$

Q4.09.1 MCQ 2009 1M

The divergence of the vector field $3xz\hat{i} + 2xy\hat{j} - yz^2\hat{k}$ at a point $(1, 1, 1)$ is equal to

A. 7 **B.** 4

C. 3 **D.** 0

Q4.09.2 MCQ 2009 2M

A path AB in the form of one quarter of a circle of unit radius is shown in the figure. Integration of $(x + y)^2$ on path AB traversed in a counter-clockwise sense is

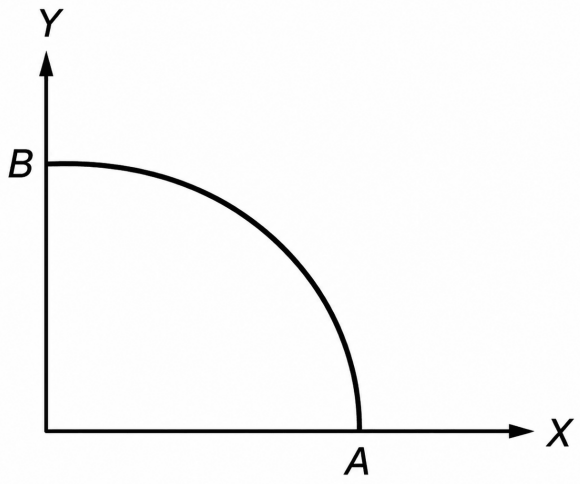


Fig. Q4.09.2

A. $\frac{\pi}{2} - 1$ **B.** $\frac{\pi}{2} + 1$

C. $\frac{\pi}{2}$ **D.** 1

Q4.08.1 MCQ 2008 2M

The directional derivative of the scalar function $f(x, y, z) = x^2 + 2y^2 + z$ at the point $P = (1, 1, 2)$ in the direction of the vector $\vec{a} = 3\hat{i} - 4\hat{j}$ is

A. -4 **B.** -2

C. -1 **D.** 1

Q4.07.1 MCQ 2007 2M

The area of a triangle formed by the tips of vectors $\overline{a}$, $\overline{b}$ and $\overline{c}$ is

A. $\frac{1}{2}(\overline{a} - \overline{b}) \cdot (\overline{a} - \overline{c})$ **B.** $\frac{1}{2}\left|(\overline{a} - \overline{b}) \times (\overline{a} - \overline{c})\right|$

C. $\frac{1}{2}\left|\overline{a} \times \overline{b} \times \overline{c}\right|$ **D.** $\frac{1}{2}(\overline{a} \times \overline{b}) \cdot \overline{c}$

Q4.05.1 MCQ 2005 2M

The line integral $\int \vec{V} \cdot d\vec{r}$ of the vector $\vec{V}(\vec{r}) = 2xyz\hat{i} + x^2z\hat{j} + x^2y\hat{k}$ from the origin to the point $P(1, 1, 1)$

A. is 1

B. is zero

C. is -1

D. cannot be determined without specifying path

Q4.05.2 MCQ 2005 1M

By a change of variables

$$x(u, v) = uv, \qquad y(u, v) = \frac{v}{u}$$

in a double integral, the integrand $f(x, y)$ changes to

$$f\left(uv, \frac{v}{u}\right)\phi(u, v).$$

Then, $\phi(u, v)$ is

A. $2\frac{v}{u}$ **B.** $2uv$

C. v^2 **D.** 1

Q4.04.1 MCQ 2004 1M

The angle between two unit-magnitude coplanar vectors $P(0.866, 0.500, 0)$ and $Q(0.259, 0.966, 0)$ will be

A. $0°$ **B.** $30°$

C. $45°$ **D.** $60°$

Q4.99.1 MCQ 1999 1M

For the function $\phi = ax^2y - y^3$ to represent the velocity potential of an ideal fluid, $\nabla^2\phi$ should be equal to zero. In that case, the value of 'a' has to be:

A. -1 **B.** 1

C. -3 **D.** 3

Q4.95.1 MCQ 1995 1M

If $\overline{V}$ is a differentiable vector function and f is sufficiently differentiable scalar function then curl$(f\overline{V})$ = _____.

A. (grad f) $\times \overline{V}$ + (f curl $\overline{V}$)

B. $\overline{o}$

C. f curl $\overline{V}$

D. (grad f) $\times \overline{V}$

Civil Engineering

Q4.26.1 MCQ 2026 2M

Vector field $\vec{V}$ is defined as

$$\vec{V} = 3x^2yz\,\hat{i} - 5xy\,\hat{j} + 6yz^2\,\hat{k}$$

The curl of $\vec{V}$ at point $(2, -1, 1)$ is

A. $6\hat{i} - 12\hat{j} - 7\hat{k}$

B. $-12\hat{i} - 10\hat{j} - 12\hat{k}$

C. -34

D. $\begin{bmatrix} -12 & 12 & -12 \\ 5 & -10 & 0 \\ 0 & 6 & -12 \end{bmatrix}$

Q4.25.1 MSQ 2025 1M

Consider a velocity vector, $\vec{V}$ in (x, y, z) coordinates given below. Pick one or more CORRECT statement(s) from the choices given below:

$$\vec{V} = u\hat{x} + v\hat{y}$$

☐ **A.** z-component of Curl of velocity;

$$\nabla \times \vec{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\hat{z}$$

☐ **B.** z-component of Curl of velocity;

$$\nabla \times \vec{V} = \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)\hat{z}$$

☐ **C.** Divergence of velocity;

$$\nabla \cdot \vec{V} = \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$$

☐ **D.** Divergence of velocity;

$$\nabla \cdot \vec{V} = \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$$

Q4.24.1 MSQ 2024 2M ○○○

Three vectors $\vec{p}$, $\vec{q}$, and $\vec{r}$ are given as

$$\vec{p} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{q} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

Which of the following is/are CORRECT?

☐ **A.** $\vec{p} \times (\vec{q} \times \vec{r}) = (\vec{p} \times \vec{q}) \times \vec{r}$

☐ **B.** $\vec{r} \cdot (\vec{p} \times \vec{q}) = (\vec{q} \times \vec{p}) \cdot \vec{r}$

☐ **C.** $\vec{p} \times (\vec{q} \times \vec{r}) + \vec{q} \times (\vec{r} \times \vec{p}) + \vec{r} \times (\vec{p} \times \vec{q}) = \vec{0}$

☐ **D.** $\vec{p} \times (\vec{q} \times \vec{r}) = (\vec{p} \cdot \vec{r})\vec{q} - (\vec{p} \cdot \vec{q})\vec{r}$

Q4.24.2 MCQ 2024 2M ○○○

A vector field $\vec{p}$ and a scalar field r are given by

$$\vec{p} = (2x^2 - 3xy + z^2)\hat{i} + (2y^2 - 3yz + x^2)\hat{j} + (2z^2 - 3zx + x^2)\hat{k}$$

$$r = 6x^2 + 4y^2 - z^2 - 9xyz - 2xy + 3xz - yz$$

Consider the statement P and Q:
P : Curl of the gradient of the scalar field r is a null vector.
Q : Divergence of curl of the vector field $\vec{p}$ is zero.
Which one of the following options is CORRECT?

A. Both P and Q are TRUE.

B. Both P and Q are FALSE

C. P is TRUE and Q FALSE.

D. P is FALSE and Q is TRUE.

Q4.23.1 MCQ 2023 1M ○○○

Let ϕ be a scalar field, and u be a vector field.
Which of the following identities is true for $\text{div}(\phi u)$?

A. $\text{div}(\phi u) = \phi\,\text{div}(u) + u \cdot \text{grad}(\phi)$

B. $\text{div}(\phi u) = \phi\,\text{div}(u) + u \times \text{grad}(\phi)$

C. $\text{div}(\phi u) = \phi\,\text{grad}(u) + u \cdot \text{grad}(\phi)$

D. $\text{div}(\phi u) = \phi\,\text{grad}(u) + u \times \text{grad}(\phi)$

Q4.21.1 MCQ 2021 1M ○○○

The unit normal vector to the surface $X^2 + Y^2 + Z^2 - 48 = 0$ at the point $(4, 4, 4)$ is

A. $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ **B.** $\frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}$

C. $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ **D.** $\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}$

Q4.21.2 NAT 2021 2M ○○○

A function is defined in Cartesian coordinate system as $f(x, y) = xe^y$. The value of the directional derivative of the function (in integer) at the point $(2, 0)$ along the direction of the straight line segment from point $(2, 0)$ to point $\left(\frac{1}{2}, 2\right)$ is _____.

Q4.20.1 NAT 2020 2M ○○○

If C represents a line segment between $(0, 0, 0)$ and $(1, 1, 1)$ in Cartesian coordinate system, the value (expressed as integer) of the line integral

$$\int_C [(y + z)\,dx + (x + z)\,dy + (x + y)\,dz]$$

is _____.

Q4.18.1 NAT 2018 2M ○○○

The value (up to two decimal places) of a line integral $\int_C \vec{F}(\vec{r}) \cdot d\vec{r}$, for $\vec{F}(\vec{r}) = x^2\hat{i} + y^2\hat{j}$ along C which is a straight line joining $(0, 0)$ to $(1, 1)$ is _____.

Q4.15.1 NAT 2015 2M ○○○

The directional derivative of the field

$$u(x, y, z) = x^2 - 3yz$$

in the direction of the vector $(\hat{i} + \hat{j} - 2\hat{k})$ at point $(2, -1, 4)$ is _____.

Q4.12.1 MCQ 2012 2M

For the parallelogram $OPQR$ shown in the sketch, $\overrightarrow{OP} = a\hat{i} + b\hat{j}$ and $\overrightarrow{OR} = c\hat{i} + d\hat{j}$. The area of the parallelogram is

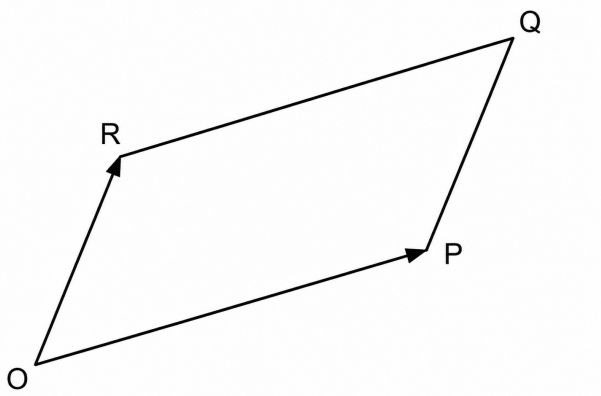


Fig. Q4.12.1

A. $ad - bc$ **B.** $ac + bd$

C. $ad + bc$ **D.** $ab - cd$

Q4.11.1 MCQ 2011 1M

If $\vec{a}$ and $\vec{b}$ are two arbitrary vectors with magnitudes a and b, respectively, $\left|\vec{a} \times \vec{b}\right|^2$ will be equal to

A. $a^2b^2 - (\vec{a} \cdot \vec{b})^2$ **B.** $ab - \vec{a} \cdot \vec{b}$

C. $a^2b^2 + (\vec{a} \cdot \vec{b})^2$ **D.** $ab + \vec{a} \cdot \vec{b}$

Q4.09.1 MCQ 2009 2M

For a scalar function $f(x, y, z) = x^2 + 3y^2 + 2z^2$, the directional derivative at the point $P(1, 2, -1)$ in the direction of a vector $\hat{i} - \hat{j} + 2\hat{k}$ is

A. -18 **B.** $-3\sqrt{6}$

C. $3\sqrt{6}$ **D.** 18

Q4.09.2 MCQ 2009 1M

For a scalar function $f(x, y, z) = x^2 + 3y^2 + 2z^2$, the gradient at the point $P(1, 2, -1)$ is

A. $2\hat{i} + 6\hat{j} + 4\hat{k}$ **B.** $2\hat{i} + 12\hat{j} - 4\hat{k}$

C. $2\hat{i} + 12\hat{j} + 4\hat{k}$ **D.** $\sqrt{56}$

Q4.06.1 MCQ 2006 2M

The directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at the point $P(2, 1, 3)$ in the direction of the vector $\vec{a} = \hat{i} - 2\hat{k}$ is

A. -2.785 **B.** -2.145

C. -1.789 **D.** 1.000

Q4.05.1 MCQ 2005 2M

Value of the integral $\oint_C (xydy - y^2dx)$, where, C is the square cut from the first quadrant by the lines $x = 1$ and $y = 1$ will be (Use Green's theorem to change the line integral into double integral)

A. $\frac{1}{2}$ **B.** 1

C. $\frac{3}{2}$ **D.** $\frac{5}{3}$

Q4.03.1 MCQ 2003 1M

If P, Q and R are three points having coordinates $(3, -2, -1)$, $(1, 3, 4)$, $(2, 1, -2)$ in XYZ space, then the distance from point P to plane OQR (O being the origin of the coordinate system) is given by

A. 3 **B.** 5

C. 7 **D.** 9

Chemical Engineering

Q4.26.1 MCQ 2026 1M

A two-dimensional temperature profile is given as $T = 2x^2 + 3xy + y^2$. If $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y directions, respectively, which one of the following is the directional derivative of T at the location $x = 2, y = 2$?

A. $8\hat{i} + 12\hat{j}$ **B.** $14\hat{i} + 10\hat{j}$

C. $8\hat{i} - 12\hat{j}$ **D.** $14\hat{i} - 10\hat{j}$

Q4.26.2 MCQ 2026 2M

Consider a flow with the following velocity field

$$\vec{V} = (x + y)\hat{i} + (y + z)\hat{j} + (z + x)\hat{k}$$

where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vectors in the x, y and z directions, respectively. Which one of the following is CORRECT?

A. The flow is incompressible and irrotational.

B. The flow is incompressible and rotational.

C. The flow is compressible and rotational.

D. The flow is compressible and irrotational.

Q4.25.1 MCQ 2025 2M

Consider a Cartesian coordinate system with orthogonal unit basis vectors $\hat{i}, \hat{j}$ defined over a domain: $x, y \in [0, 1]$. Choose the condition for which the divergence of the vector field $v = ax\hat{i} - by\hat{j}$ is zero.

A. $a - b = 0$ **B.** $a < b$

C. $a > b$ **D.** $a + b = 0$

Q4.24.1 NAT 2024 2M

Consider the line integral $\int_C F(r) \cdot dr$, with $F(r) = x\hat{i} + y\hat{j} + z\hat{k}$, where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors in the (x, y, z) Cartesian coordinate system. The path C is given by $r(t) = \cos(t)\hat{i} + \sin(t)\hat{j} + t\hat{k}$, where $0 \le t \le \pi$. The value of the integral, rounded off to 2 decimal places, is _____.

Q4.23.1 MCQ 2023 1M

For a two-dimensional plane, the unit vectors, $(\hat{e}_r, \hat{e}_\theta)$ of the polar coordinate system and $(\hat{i}, \hat{j})$ of the Cartesian coordinate system, are related by the following two equations.

$$\hat{e}_r = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$$

$$\hat{e}_\theta = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$

Which one of the following is the CORRECT value of $\frac{\partial(\hat{e}_r + \hat{e}_\theta)}{\partial\theta}$?

A. 1 **B.** $\hat{e}_\theta$

C. $\hat{e}_r + \hat{e}_\theta$ **D.** $-\hat{e}_r + \hat{e}_\theta$

Q4.22.1 NAT 2022 2M

Consider a sphere of radius 4, centered at the origin, with outward unit normal $\hat{n}$ on its surface S. The value of the surface integral

$$\iint_S \left(\frac{2x\hat{i} + 3y\hat{j} + 4z\hat{k}}{4\pi}\right) \cdot \hat{n}\, dA$$

is _____.

(rounded off to one decimal place)

Production & Industrial Engineering

Q4.26.1 NAT 2026 1M

Consider the line integral

$$\int_C (2x\,dx + 2y\,dy + 2z\,dz)$$

where C is a semi-circle in the $z = 0$ plane with start point at $(0, 0, 0)$ and end point at $(1, 0, 0)$.

The value of the line integral is _____ (in integer).

Q4.25.1 NAT 2025 2M

If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the orthogonal unit vectors in Cartesian x-y-z coordinate system, the rate of change of the function $f(x, y, z) = x^2 + 2y^2 + z$ at point $(1, 1, 1)$ in the direction of $3\hat{i} + 4\hat{k}$ is _____. (Answer in integer)

Q4.24.1 MCQ 2024 1M

A vector field is given as $F(x, y) = (100x + 100y)\hat{i} + (-50x + 200y)\hat{j}$, where $\hat{i}$ and $\hat{j}$ are the unit vectors along the x and y axes in the Cartesian frame, respectively. Then the value of

$$\oint_C F(x, y) \cdot dl$$

where $dl = dx\hat{i} + dy\hat{j}$ is an elemental path taken over an anticlockwise circular contour C of radius $r = 2$ is

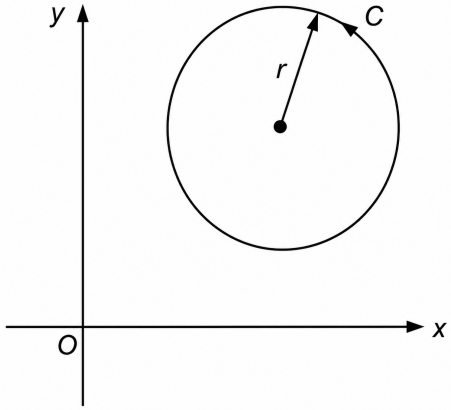


Fig. Q4.24.1

A. -100π **B.** -800π

C. -400π **D.** 400π

Q4.22.1 MCQ 2022 1M

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors, the vector triple product $(\vec{a} \times \vec{b}) \times \vec{c}$ is given by

A. $(\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b}$ **B.** $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$

C. $(\vec{b} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{c})\vec{b}$ **D.** $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

Q4.18.1 NAT 2018 1M

The value of surface integral

$$\iint_S (9x\hat{i} - 2y\hat{j} - z\hat{k}) \cdot \hat{n}\, dS$$

over the surface of sphere S, $x^2 + y^2 + z^2 = 9$ where $\hat{n}$ is unit outward normal to surface element dS is _____.

Q4.17.1 MCQ 2017 1M

For two non-zero vectors $\vec{A}$ and $\vec{B}$, if $\vec{A} + \vec{B}$ is perpendicular to $\vec{A} - \vec{B}$ then,

A. Magnitude of $\vec{A}$ is twice magnitude of $\vec{B}$.

B. Magnitude of $\vec{A}$ is half the magnitude of $\vec{B}$.

C. $\vec{A}$ and $\vec{B}$ cannot be orthogonal.

D. The magnitudes of $\vec{A}$ and $\vec{B}$ are equal.

Q4.14.1 MCQ 2014 2M

If $\phi = 2x^3y^2z^4$ then $\nabla^2\phi$ is

A. $12xy^2z^4 + 4x^2z^2 + 20x^3y^2z^3$

B. $2x^2y^2z + 4x^3z^4 + 24x^3y^2z^2$

C. $12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2$

D. $4xyz^2 + 4x^2z^2 + 24x^3y^2z^2$

Q4.14.2 MCQ 2014 1M

Directional derivative of $\phi = 2xz - y^2$ at the point $(1, 3, 2)$ becomes maximum in the direction of:

A. $4\hat{i} + 2\hat{j} - 3\hat{k}$ **B.** $4\hat{i} - 6\hat{j} + 2\hat{k}$

C. $2\hat{i} - 6\hat{j} + 2\hat{k}$ **D.** $4\hat{i} - 6\hat{j} - 2\hat{k}$

Q4.11.1 MCQ 2011 1M

If $A(0, 4, 3)$, $B(0, 0, 0)$ and $C(3, 0, 4)$ are three points defined in x, y, z co-ordinate system, then which of the following vector is perpendicular to both vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$.

A. $16\hat{i} + 9\hat{j} - 12\hat{k}$ **B.** $16\hat{i} - 9\hat{j} + 12\hat{k}$

C. $16\hat{i} - 9\hat{j} - 12\hat{k}$ **D.** $16\hat{i} + 9\hat{j} + 12\hat{k}$

Q4.11.2 MCQ 2011 1M

The line integral

$$\int_{P_1}^{P_2} (y\, dx + x\, dy)$$

for $P_1(x_1, y_1)$ to $P_2(x_2, y_2)$ along the semicircle P_1, P_2 shown in the figure is

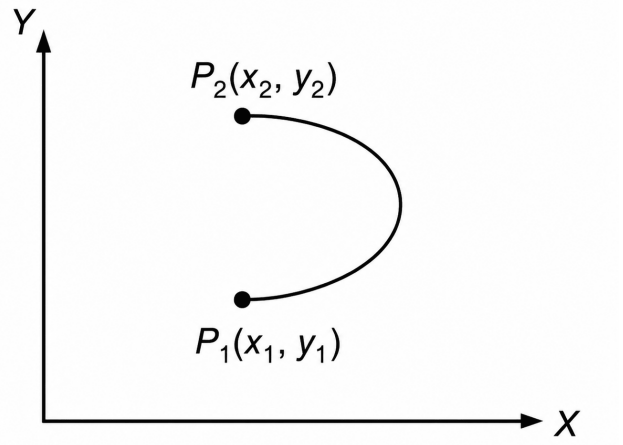


Fig. Q4.11.2

A. $x_2y_2 - x_1y_1$ **B.** $(y_2^2 - y_1^2) + (x_2^2 - x_1^2)$

C. $(x_2 - x_1)(y_2 - y_1)$ **D.** $(y_2 - y_1)^2 + (x_2 - x_1)^2$

Q4.08.1 MCQ 2008 1M

If $\vec{r}$ is the position vector of any point on a closed surface S that encloses volume V then $\iint_S \vec{r} \cdot d\vec{S}$ is equal to

A. $\frac{1}{2}V$ **B.** V

C. $2V$ **D.** $3V$

Q4.07.1 MCQ 2007 2M

Divergence of vector field $\vec{V}(x, y, z) = -(x \cos xy + y)\hat{i} + (y \cos xy)\hat{j} + [(\sin z^2) + x^2 + y^2]\hat{k}$ is

A. $2z \cos z^2$ **B.** $\sin xy + 2z \cos z^2$

C. $x \sin xy - \cos z$ **D.** None of these

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q4.26.1 D	Q4.24.1 A,C	Q4.24.2 A,B,D	Q4.23.1 C	Q4.23.2 B	Q4.23.3 B
Q4.22.1 A	Q4.21.1 A	Q4.20.1 C	Q4.19.1 C	Q4.17.1 -11	Q4.17.2 B
Q4.17.3 54 to 55	Q4.16.1 20	Q4.16.2 0	Q4.15.1 A	Q4.14.1 3	Q4.14.2 3
Q4.14.3 3.13 to 3.15	Q4.13.1 D	Q4.13.2 D	Q4.12.1 A	Q4.11.1 D	Q4.10.1 C
Q4.08.1 B	Q4.08.2 A	Q4.06.1 A	Q4.93.1 1		
ELECTRICAL ENGINEERING					
Q4.26.1 0	Q4.25.1 0 to 1	Q4.24.1 C	Q4.23.1 10	Q4.23.2 9 to 10	Q4.23.3 D
Q4.22.1 C	Q4.22.2 C	Q4.22.3 B	Q4.20.1 -3	Q4.20.2 B	Q4.19.1 139
Q4.18.1 0	Q4.18.2 A	Q4.16.1 4 to 5	Q4.16.2 B	Q4.15.1 0 to 1	Q4.14.1 A
Q4.14.2 B	Q4.13.1 B	Q4.13.2 D	Q4.12.1 A	Q4.11.1 B	Q4.10.1 A
Q4.09.1 D	Q4.06.1 D	Q4.05.1 C			
INSTRUMENTATION ENGINEERING					
Q4.25.1 B	Q4.20.1 C	Q4.19.1 C	Q4.18.1 D	Q4.17.1 0.65 to 0.80	Q4.16.1 D
Q4.15.1 A	Q4.14.1 1	Q4.13.1 D	Q4.12.1 A	Q4.09.1 A	
MECHANICAL ENGINEERING					
Q4.26.1 A	Q4.25.1 3	Q4.25.2 1	Q4.25.3 D	Q4.24.1 B	Q4.23.1 0
Q4.22.1 A	Q4.22.2 8 to 9	Q4.22.3 A	Q4.20.1 A	Q4.20.2 A	Q4.20.3 6
Q4.19.1 B	Q4.19.2 C	Q4.19.3 C	Q4.18.1 C	Q4.18.2 C	Q4.18.3 C
Q4.17.1 226	Q4.17.2 0	Q4.17.3 C	Q4.17.4 B	Q4.16.1 16	Q4.16.2 726
Q4.15.1 216	Q4.15.2 1 to 2	Q4.15.3 A	Q4.15.4 1.86	Q4.14.1 C	Q4.14.2 A

Q4.14.3 B	Q4.14.4 C	Q4.13.1 A	Q4.12.1 A	Q4.10.1 D	Q4.09.1 C
Q4.09.2 B	Q4.08.1 B	Q4.07.1 B	Q4.05.1 A	Q4.05.2 A	Q4.04.1 C
Q4.99.1 D	Q4.95.1 A				
CIVIL ENGINEERING					
Q4.26.1 A	Q4.25.1 A,C	Q4.24.1 B,C,D	Q4.24.2 A	Q4.23.1 A	Q4.21.1 A
Q4.21.2 1	Q4.20.1 3	Q4.18.1 0 to 1	Q4.15.1 -5.72 to -5.70	Q4.12.1 A	Q4.11.1 A
Q4.09.1 B	Q4.09.2 B	Q4.06.1 C	Q4.05.1 C	Q4.03.1 A	
CHEMICAL ENGINEERING					
Q4.26.1 B	Q4.26.2 C	Q4.25.1 A	Q4.24.1 4.5 to 5	Q4.23.1 D	Q4.22.1 192
PRODUCTION & INDUSTRIAL ENGINEERING					
Q4.26.1 1	Q4.25.1 2	Q4.24.1 *	Q4.22.1 B	Q4.18.1 677 to 679	Q4.17.1 D
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